

Chapter 3

Data and Methodology

3.1 Research design and evaluation criteria

This work is a predictive-modelling study: we ask whether label-free *distances* measurable in advance (X) predict a frozen tracker’s zero-shot performance ($Y = \text{pDetA}/\text{pAssA}$), and which factors govern that transfer for *finding* versus *following* an animal. Because the target is a bounded score and the observations are grouped by species and location, the analysis is a weighted regression with group-aware, out-of-sample validation rather than a tuned tracker.

Definition of success and the decision rule. The intended deliverable is a *validated, out-of-sample* predictor of $\text{pDetA}/\text{pAssA}$ from label-free distances, with honest confidence intervals. A distance is taken to predict transfer only if it satisfies **both** criteria: (i) its coefficient’s group-cluster-bootstrap 95% confidence interval excludes zero, *and* (ii) the grouped cross-validated model beats a mean-predictor baseline by a margin that is significant under a paired group-bootstrap. **H1** (detection $\leftarrow$ species novelty) and **H2** (association $\leftarrow$ environment difficulty) are supported only when their respective distances meet this bar on the split that isolates them; if neither does under these deliberately powerful tests, we conclude **H0** — distance explains little variance — which is itself a valid result and motivates the representational-probing direction of Chapter 5.

3.2 Dataset, splits, and the analysis unit

We use SA-FARI [20] throughout. In its annotations, each test is a *probe* — a (video, text-prompt) pair; the prompt resolves the animal’s species and its seven-level taxonomy, while each video’s location and timestamp give place and time. We aggregate scores to a *cell* — a (species, location, time) group, keyed by a stable numeric species identifier — and attach support counts (n_{frames} , n_{masklets} , n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection; the one exception is the stratified per-species cap used to make the large pooled pass feasible, under which hard negatives are set aside from inference and analysed instead by the separate false-positive pass (Section 3.7).

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by *location* but shares species across train and test (Section 2.6); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM3 is frozen and zero-shot on all of SA-FARI, the split only defines our *reference distribution*, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).

Pooled species are partitioned by a *leave-one-species-out* view, so a held-out species’ distance is to the nearest *other* species; species-distance varies while the environment stays largely familiar, isolating H1. Validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).

The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the Phase-1 score-sanity check.

Split A is a *constructed* hold-out — legitimate because the model is frozen, and reported as such. At the scale we work with, the location split (Split B) yields 1,447 scored cells over 113 taxonomy-bearing category identities (99 species) and 92 unseen test locations: 346 are *positive* cells — the queried species is present, so $\text{pDetA}/\text{pAssA}$ are defined — and 1,101 are hard-negative cells, drawn from 1,486 hard-negative probes. The species hold-out (Split A) pools the present species of both official splits into 1,025 scored cells across the 99 species.

3.3 Measuring transfer

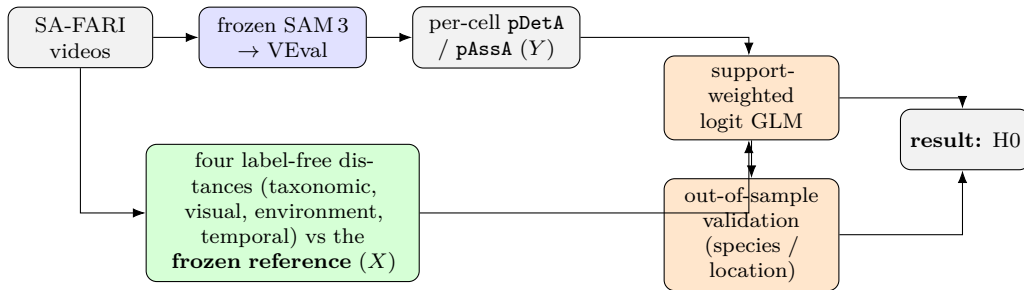


Figure 3.1: The pipeline. *Top:* frozen SAM3 is run over the videos and scored by the official evaluator into per-cell detection/association scores (the target Y). *Bottom:* four label-free distances are computed from the videos and ground-truth masks against a frozen reference set (the predictors X). A support-weighted logit GLM relates X to Y and is validated out of sample by holding out whole species and whole locations; SAM3 is never trained.

SAM3 [5] is run frozen over the probe videos (species-specific prompts as the primary condition, a generic “animal” prompt for robustness) and scored with the official **VEval** evaluator — the metric is never re-implemented — giving per-cell $\text{pDetA}/\text{pAssA}/\text{pHOTA}$ [13] with their support. A single inference pass over the union of Split A and Split B probes serves both experiments; predictions are stored per (video, species) so that the several species-prompts issued on one video do not collide, and the harness is resumable at per-video granularity and never updates a weight. We aggregate two ways and keep them distinct: the *micro* dataset-level **VEval** score (the published SA-FARI operating point, ≈ 0.65 mask-**pHOTA**) and the per-cell support-weighted *macro* mean over positive cells that is our modelling target. Because the macro mean is a positives-only per-cell average it sits lower (≈ 0.53), and every Gate-1 number we quote is the macro value, labelled as such.

No retraining; validity of the frozen protocol. It bears stating plainly, as it is easily misread: **SAM3 is never retrained or fine-tuned in this work.** The single object we fit is the small support-weighted logistic regression that maps the label-free distances X onto the observed scores Y ; the tracker itself is the public checkpoint, run once, its weights untouched. Reusing that one inference across both splits is therefore sound rather than a leak: because the model is frozen, a cell’s score does not depend on which reference partition we later overlay to compute its distances. This also fixes the benchmark’s status here. SA-FARI is a held-out, *zero-shot* benchmark for the released SAM3: the SAM3 report [5] does not list SA-FARI

among its training sources, and the SA-FARI paper [20] reports the released checkpoint zero-shot (the ≈ 0.65 mask-pHOTA ballpark we reproduce at Gate 1) *alongside* separately fine-tuned rows whose gain we deliberately forgo. We can therefore verify non-membership at the *dataset* level; we cannot verify it frame-by-frame, since SAM 3’s full pretraining corpus is undisclosed. We do not claim otherwise: every distance is measured against the SA-FARI *train reference*, not against SAM 3’s true (unknown) pretraining distribution, and the familiarity proxy (Section 3.5, planned) is the hedge for exactly this gap.

3.4 The frozen reference (leakage firewall)

For each split the reference (“seen”) species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a feature — verified by a unit test that swaps the reference and checks the distances move as expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and *report* any residual species overlap honestly rather than asserting it away.

3.5 Label-free distance features

All distances are label-free for the target and computed against the frozen reference.

3.5.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe’s species to the nearest reference species (shared genus = 1, family = 2, order = 3, ...); the leave-species-out reference keeps this non-trivial for shared species.

3.5.2 Visual distance

Each species is fingerprinted by a *prototype* — the re-normalised mean DINOv2 [16] embedding of its *ground-truth-mask-cropped* animals — and the visual distance is the cosine gap from a probe species’ prototype to that of the nearest *other* reference species (leave-one-species-out on Split A), with a CLIP [17] encoder as a robustness check. Using the dataset’s ground-truth masks means this feature needs no SAM 3.

3.5.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus an interpretable day/night–infrared flag (from per-channel colour statistics of the frame) and a clutter score given by the variance of a Laplacian over the masked background — a cheap measure of local edge/texture density, not a colour statistic.

3.5.4 Temporal gap

The minimum absolute year gap between a cell’s capture year (from the per-video creation-datetime metadata) and the nearest reference year. On the location split the timestamp is present for few cells and near-constant among them, so the temporal axis is reported there as *untested* rather than as a tested null.

3.5.5 Animal size (confound control)

Mean log ground-truth mask area per species. This is not a hypothesis variable but a nuisance covariate, included to separate genuine visual novelty from the fact that visually-distinctive



Figure 3.2: The two image-based distances on a real camera-trap frame (a collared peccary). **(a)** the frame with its ground-truth mask (orange); **(b)** the mask-cropped animal that the *visual* distance fingerprints (background zeroed); **(c)** the same frame with the animal mean-inpainted, which the *environment* distance fingerprints. The per-video capture date (a metadata field, not the pixels) drives the *temporal* gap. None of these features runs SAM3.

species tend to be large and thus easier to segment; its role in exposing the one apparent effect as an artefact is Section 3.6.

3.5.6 SAM3 familiarity proxy

Because SAM3’s true pretraining corpus is undisclosed, every distance above is measured against the SA-FARI reference, not against what SAM3 actually saw (§3.9 caveat). The familiarity proxy hedges this by reading familiarity *from the model itself*: how separable each species is in SAM3’s *own* vision-encoder feature space. The same mask-cropped animals used for the visual distance are pushed through SAM3’s image encoder (`get_vision_features`, mean-pooled over the patch tokens to a 1024-d vector), and each species is scored three ways — *silhouette* (own-cluster compactness vs the nearest other species), *nearest-prototype* (the visual distance recomputed with SAM3’s encoder rather than DINOv2), and *Mahalanobis* typicality against the reference feature Gaussian. Like the other distances it is **before-running** (an image property, not a function of SAM3’s tracking output — so, unlike the after-running confidence estimator, non-circular) and **label-free**. Each metric is validated at the exact leave-species-out bar (Bonferroni over the three), size-controlled, and put in a head-to-head against the DINOv2 visual distance: the decisive question is whether it adds any out-of-sample gain over `visual_distance + size` (§4.8).

3.6 Statistical modelling and hypothesis testing

3.6.1 Per-target regression

Each bounded target (`pDetA`, `pAssA`) is modelled *separately* with a support-weighted logit-link (fractional-logit) regression over the standardised distances plus the scene covariates and a $\log(n_{\text{frames}})$ covariate, so that “rare” is never mistaken for “far”. Detection and association are analysed apart because their decomposition is the intellectual core of the project. In practice association is a *near-degenerate* target on this data — most cells contain a single object, so `pAssA` tracks `pDetA` almost exactly — and it is therefore reported as such rather than as an independently powered test (Chapter 5).

3.6.2 Honest, group-aware inference

Naive model confidence intervals are invalid here for two reasons: support-weighting by frame count inflates the effective sample size, and the distances are constant within a species or a location, so the hundreds of cells are really only tens of independent groups (pseudo-replication).

We therefore report *group-cluster-bootstrap* intervals — resampling whole species and whole locations, refitting, and taking percentile intervals — as the honest confidence interval for every coefficient.

3.6.3 Out-of-sample validation and the significance test

The fitted models are validated *out of sample* by holding out whole groups (leave-species-out on Split A, leave-location-out on Split B) and predicting the held-out group. The out-of-sample gain over a mean-predictor baseline is tested with a *paired group-bootstrap* that resamples whole held-out groups, yielding a confidence interval and a one-sided *p*-value on the gain — the criterion (ii) of Section 3.1.

3.6.4 Variance attribution and confound control

Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and raw correlations are reported as a model-free cross-check. Where a distance reaches significance, we refit with the animal-size covariate to test whether the effect is a size confound rather than novelty.

3.6.5 The intrinsic-difficulty pivot and the nested decomposition

Because the sole apparent effect proved to be a size confound, we ask a complementary question: is transfer governed by the intrinsic *difficulty* of the target imagery rather than its *novelty*? We promote the label-free scene properties already measured for the environment axis — low-light/IR (the low-light fraction), clutter, and the night/IR flag — from nuisance covariates to predictors of interest, and test them against the same out-of-sample bar. To separate a genuine difficulty effect from the size confound, we fit a *nested* sequence of models — support only; size only; difficulty only (no size); difficulty plus size — and compare their out-of-sample gains, so that any apparent difficulty effect that is really size is exposed as such (Chapter 4, Table 4.4). We also report each scene property’s *cell-level* correlation alongside its location-level value, because aggregation inflates correlations (the ecological-correlation fallacy).

Detecting a signal, not asserting its absence. These tests are demonstrably able to detect a real effect: the animal-size covariate not only has an interval excluding zero but *validates out of sample* at the same whole-group hold-out bar the distances fail (Chapter 4), and the estimation and cross-validation machinery is unit-tested to recover planted effects on synthetic data. A null under a test that fires for size is therefore evidence of *absence of signal*, not of *absence of power*.

3.6.6 Design-robustness checks

To show the null is not an artefact of a single design choice, we re-run the entire hold-out procedure with one component of the pipeline swapped at a time: the visual encoder (DI-NOv2 → CLIP), the crop (mask-cropped animal → a whole-frame bounding box that keeps the background), the prompt (species-specific → a generic “animal”), and the visual-distance functional itself (nearest-prototype cosine → a distributional Fréchet/MMD distance between the target and reference embedding sets, after AutoEval [7]). Each variant is fitted and validated at the identical whole-group hold-out bar of Section 3.1, with significance judged after a Bonferroni correction over the variant family; the outcomes are reported in Chapter 4 (Table 4.11).

3.7 Detection precision: false positives on hard negatives

The pDetA model conditions on the animal being present, so it measures recall-given-present, not hallucination. On hard negatives (the queried species is absent) a correct run returns no

masklet; we read the prediction directly, flag any returned masklet above a score threshold as a false positive, attach each species’ taxonomic and visual distance, and test — at the species level, to avoid pseudo-replication — whether the hallucination rate rises with novelty.

3.8 Independent replication on external datasets

Every SA-FARI result reuses one frozen inference pass on one dataset, so the two splits are near-orthogonal but not statistically independent. To test whether the before-running null is a property of the transfer problem rather than of SA-FARI, we re-run the *entire* pipeline — unchanged frozen SAM3, the same official scorer, the same distances, and the same leave-species-out cross-validation and paired group bootstrap — on two independent tracking datasets. The abstraction boundary that makes this cheap is the annotation schema, not the loader: an offline adapter converts each dataset into the SA-Co JSON the loader and evaluator already read, so nothing downstream changes. Only the small logit GLM is ever fitted; SAM3 is frozen throughout.

MammAlps (box-only, small). An Alpine camera-trap tracker [8]: 5 species over 3 sites / 9 cameras, box-only ground truth. The adapter emits each box as a filled-rectangle RLE (so the mask-cropped visual, size and environment features run) together with the native box, and the scorer reads box-HOTA rather than mask-HOTA (against a filled rectangle a tight predicted mask has near-zero IoU, so box-IoU is the honest metric). The 30 fps clips are subsampled to 3 fps and capped per (species, camera) cell; the per-camera site metadata supports both leave-species-out and leave-camera-out. Only three of the four distances are testable (temporal is degenerate over the six-week window; taxonomic is near-constant across five species).

BURST (mask-native, many-species). The animal subset of the BURST validation split [1] (built on TAO [6]): 41 animal species across 132 video $\times$ class cells, with native COCO-RLE masks scored by mask-HOTA. The animal categories are selected from BURST’s 482 LVIS classes [10] by a WordNet filter [14] (hyponyms of *animal.n.01*) and mapped to a seven-level biological taxonomy where resolvable (25/41 species full, the rest left undefined). BURST carries no site or timestamp metadata, so the cell key is the per-video sequence and leave-species-out is the only valid grouping; the association target is genuinely non-degenerate here, so both components are tested. Masks stream from the RWTH server and TAO frames from the Hugging Face mirror by HTTP range, so only the annotated frames of the capped animal videos are fetched.

Interpreting the replications. Both datasets are held to the exact same out-of-sample bar as SA-FARI, and both include the animal-size confound control and the after-running confidence features as positive controls — a fired control shows the test is alive at that sample size, and its failure to fire (as on the 20-cell MammAlps) marks the design as underpowered rather than the hypothesis as confirmed. We report convergence across datasets, not independent proof: BURST detects only a large between-species predictor, so it cannot exclude a small before-running distance effect (§4.10).

3.9 Reproducibility, implementation, and constraints

The model is frozen and scored by the official evaluator; the seen set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. The pipeline is config-driven (every constraint is a toggle, so an ablation flips config, not code) and covered by hermetic tests; every reported number is regenerated from the repository, and split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to

end. The engineering safeguards that keep the measurement honest — memory-lean feature assembly, per-(video, species) prediction keying, trimming the ground truth to the scored videos before evaluation, and the size-confound control — are catalogued in the appendix, each framed as a validity safeguard.